CSE334S (UG2023) – Database Systems

Spring 26

PHASE 1

TEAM ID: 16

NAME	ID	sec	BN
ياسين أيمن السيد حراز	2301087	4	112
عبد الرحمن هاني أشرف	2300130	3	13
سما جمال أحمد	2300371	1	166
ملك ممدوح صلاح	2300428	1	175
أحمد معتز حفني	2301040	1	57
منة أسامة محمد	2300515	1	122

SOCIAL MEDIA SYSTEM DESCRIPTION

Introduction:

A social media platform is a digital system that enables users to create, share, and interact with content while connecting with others online. It allows individuals to publish posts, upload media such as photos and videos, engage through comments and likes, and discover content using features like hashtags. Users can also build networks by following other users or topics, creating a highly interactive and dynamic environment.

These platforms are designed to handle large volumes of data and complex relationships between users and content. A well-structured database is essential to efficiently manage user information, content creation, interactions, and activity tracking. Overall, social media platforms play a major role in communication, content sharing, and community building in the modern digital world.

The system consists of 7 main entities:

USERS:

The central entity of the entire system. It represents every registered account on the platform and contains the following attributes:

- A unique ***user_id*** which serves as the primary key.
- A ***username*** which acts as a unique identifier visible to others
- A ***account_info*** composite attribute that contains the following**:**
- An ***email*** **address**
- An ***photo_url***
- A ***bio***
- A ***created_at*** timestamp that records when the account was first created.

A user can create many posts, write many comments, have many login records, like many posts, like many comments, bookmark many posts, follow many users, and follow many hashtags, making USERS the most connected entity in the system with predominantly 1:M and M:N relationships branching outward from it.

POSTS:

The **POST** entity represents any piece of content published by a user on the platform. Each post is identified by:

- A unique ***post_id*** as its primary key
- A ***caption***
- A ***location*** tag
- A ***created_at*** timestamp

A post belongs to exactly one user (M:1 with USERS), can contain many photos (1:M with PHOTOS), many videos (1:M with VIDEOS), and many comments (1:M with COMMENTS). It can also be liked by many users and bookmarked by many users (M:N), and can be tagged with many hashtags (M:N with HASHTAGS).

PHOTOS:

The **PHOTOS** entity stores image content attached to posts. It is identified by:

- A unique ***photo_id*** as its primary key
- ***image_url***
- A ***created_at*** timestamp

Each photo belongs to exactly one post, forming a many-to-one (M:1) relationship with the POST entity, meaning one post can have many photos but each photo is tied to only one post.

Videos:

The **VIDEOS** entity stores video content attached to posts. It identified by:

- A unique ***video_id*** as its primary key.
- A ***video_url***
- A ***created_at*** timestamp

Just like PHOTOS, each video belongs to exactly one post, forming a many-to-one (M:1) relationship with the POST entity, where one post can have many videos but each video references only one post.

COMMENTS:

The **COMMENTS** entity captures all textual responses that users leave on posts. Each comment is identified by:

- A unique ***comment_id*** as its primary key
- ***comment_text***
- A ***created_at*** timestamp.

A comment is written by exactly one user and belongs to exactly one post, forming many-to-one (M:1) relationships with both USERS and POST. However, a comment can be liked by many users, giving it an M:N relationship with USERS through the COMMENT_LIKES associative table that has an attribute that's specific to the relationship named LIKE_COUNT.

HASHTAGS:

The **HASHTAGS** entity represents topic tags used to categorize and discover content across the platform. Each hashtag is identified by:

- A unique ***hashtag_id*** as its primary key
- The ***hashtag_name***
- A ***created_at*** timestamp.

A hashtag can be associated with many posts and followed by many users, forming M: N relationships with POST resolved through the POST_TAGS associative table respectively.

LOGIN:

The **LOGIN** entity tracks the session history of users on the platform. It records each login event with:

- *login_id* as its primary key
- A *login_time* timestamp
- A *login_ip* address

Each login record belongs to exactly one user, forming a many-to-one (M:1) relationship with the USERS entity, where one user can have many login records over time but each record is linked to only one user.

ERD:

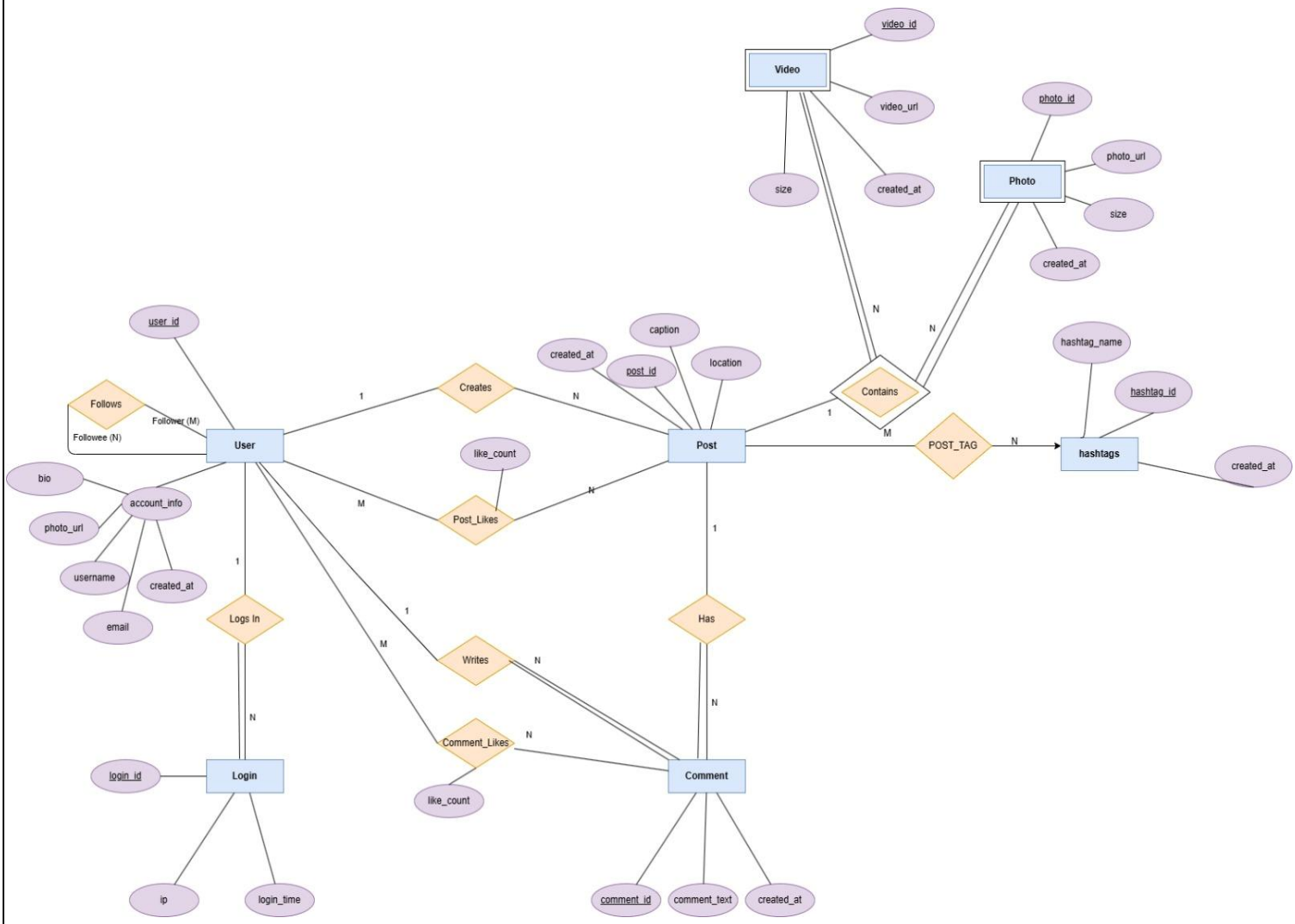
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Photo
photo_id
photo_url
size
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Contains
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1
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User
user_id
Follows
Follower (M)
Followee (N)
bio
account_info
photo_url
username
created_at
email
Creates
1
N
Post
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created_at
post_id
location
like_count
Post_Likes
M
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POST_TAG
N
hashtags
hashtag_name
hashtag_id
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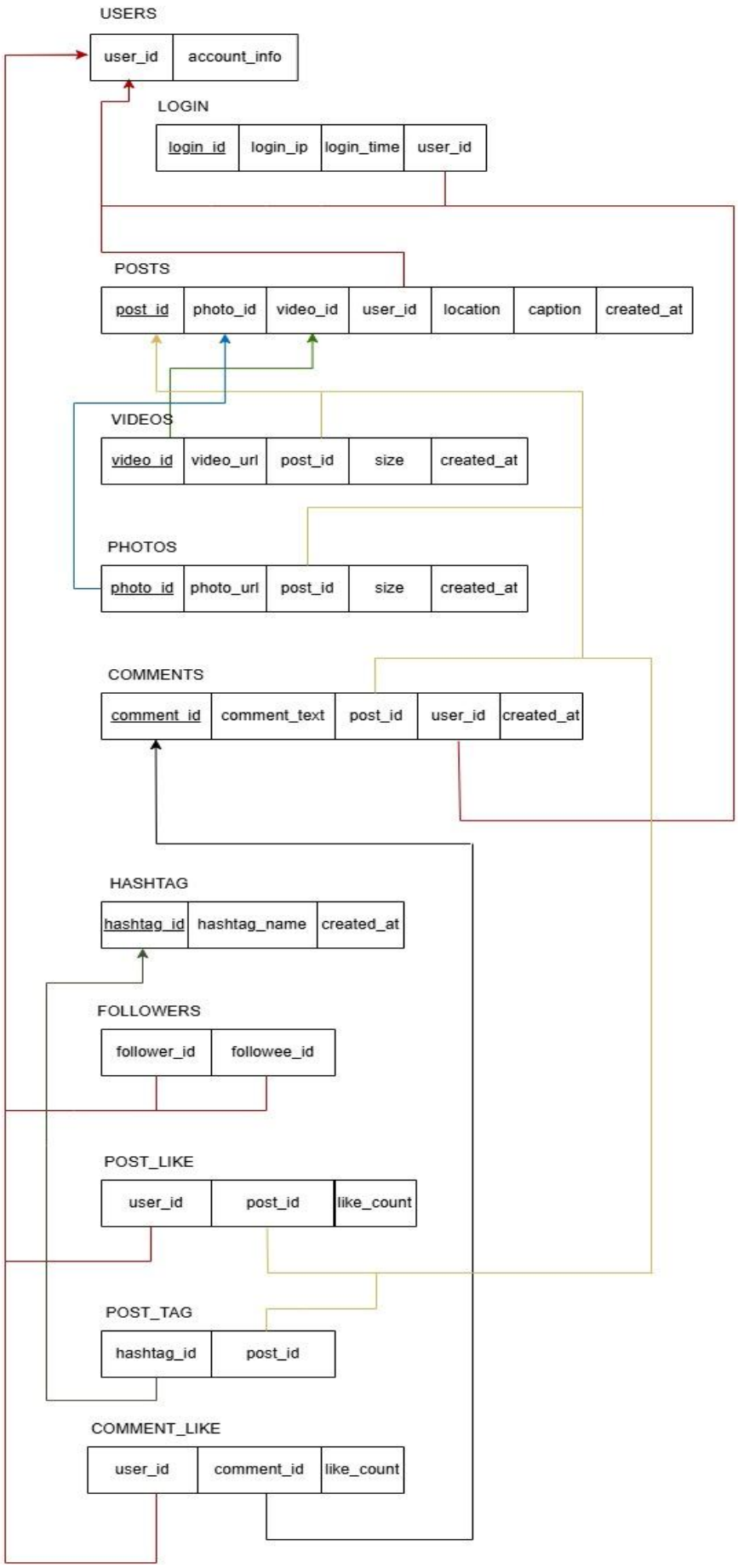
created_at


USERS
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LOGIN
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login_time
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POSTS
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COMMENTS
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FOLLOWERS
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COMMENT_LIKE
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